

EC551 - Advanced Digital Design with Verilog and FPGAs Fall 2026

Class: Tuesday and Thursday, 1:30 pm – 3:15 pm in CGS 121

Lab: Section N/A

Number of credits: 4, Prerequisites: EC311, EC413

Course Objectives

This course introduces advanced undergraduates and early-stage graduate students to advanced digital logic design by presenting the “electronic design automation” (EDA) design methodology and associated workflows. This methodology includes the steps of Design specification, Design partition, Design entry, Simulation/functional verification, Design integration and verification, Presynthesis sign-off, Synthesize and map gate-level netlist, Postsynthesis design validation, Postsynthesis timing verification, Test generation and fault simulation, Cell placement, scan chain and clock tree insertion, **cell routing, DRC, Extract parasitics**, and Design sign-off. All but the topics highlighted in RED will be covered in this course.

Content includes the use of a hardware description language (HDL; in particular Verilog) for the specification, synthesis, simulation, and exploration of principles of register transfer level (RTL) designs. Programmable logic, such as field programmable gate array (FPGA) devices, has become a major component of digital design. In this class, the students **have the opportunity** to write HDL models that can be automatically synthesized into integrated circuits using FPGAs. **Alternatively**, they can choose to write programs to implement aspects of the EDA design flow. Programming and homework exercises include writing HDL models of combinational and sequential circuits, synthesizing models, performing simulations, writing testbench modules, and synthesizing designs to an FPGA by using automatic place and route EDA tools.

The course material is broken down into four areas: Electronic Design Automation (EDA), Verilog, Reconfigurable Computing (RC), and the course Project. The topics are scheduled and organized so that these topics are woven together in lecture to provide breadth and depth in select areas. Homework (HW) and Programming exercises are created to reinforce lecture material and provide the required knowledge for the larger course project. At the conclusion of the course, the students should understand a digital EDA design flow and how to program reconfigurable computing hardware using this flow.

The course project plays a major role in the course. Students are expected to create a fully demonstrable digital system on an FPGA **OR** create a software project that demonstrates their understanding of the traditional EDA digital logic synthesis design flow. This project is done in groups of 4-5 students and should represent 15 hours of work (or more) per student per week.

Staff Information

Instructor

Name: **Prof. Douglas Densmore**

Office: **CILSE 403**

Office phone number: 617-358-6238 (email is best)

E-mail address: dougd@bu.edu (**Best way to contact me - Include EC551 in the subject line**)

Office Hours: Thursdays 10-11am in CLISE 403

Zoom Link: <https://bostonu.zoom.us/j/4602829913>

Lab Assistant

Name: TBD

E-mail address: TBD@bu.edu (**Include EC551 in the subject line**)

Office Hours: TBD TBD or by appointment

Grader

TBD

Email Address: TBD@bu.edu (**Include EC551 in the subject line**)

Course Resources

Required Textbooks

1. Author: M.D. Ciletti
Title: Advanced Digital Design with the Verilog HDL (2nd Edition)
ISBN: 0136019285

Optional Textbooks

1. Author: David R. Smith, Paul Franzon
Title: Verilog Styles for Synthesis of Digital Systems
ISBN: 0201618605
Hard to find but HIGHLY recommended
2. Author: Samir Palnitkar
Title: Verilog HDL
ISBN: 0132599708

Announcements, course material, and other useful links

Will be posted on Blackboard Learn (<http://learn.bu.edu>)

- EC551/Fall 2026

We will also use Piazza as a forum for questions on HW, programming assignments, and the class project

- <https://piazza.com/bu/fall2026/engec551/home>

Goals

To provide students with:

- An experience of how to write HDL models that can be automatically synthesized into integrated circuits using programmable hardware such as FPGAs.
- An understanding of how to take an electronic design from concept to register transfer level (RTL) verification and synthesis to final programmable device implementation.
- An experience in writing HDL models of combinational and sequential circuits, synthesizing models, performing simulations, writing test modules, and fitting designs within resource, power, and timing constraints of an FPGA by using automatic place and route CAD software.
- An experience writing software tools that perform various aspects of the EDA workflow ultimately tying the tools together into one cohesive program.

Course Outcomes

As an outcome of completing this course, students should be able to:

- Understand advanced topics in digital logic design
- Understand proven design methodologies based on standard EDA tools
- Understand the differences and similarities in hardware and software design
- Understand modern specification methods (HDL)
- Design combinational devices with a full set of EDA tools (skills)
- Understand modeling and verification with hardware description languages
- Understand synthesis with HDLs
- Understand programmable logic devices and FPGAs
- Design state machines, datapath controllers, and assorted CPUs with a full set of EDA tools
- Understand timing analysis
- Understand fault simulation and testing

Evaluation

Grading

Midterm - 10%

Final Exam - 10%

Final Project - 40%

Programs (3) - 30%; (10%, 10%, 10%)

Five Homework Assignments - 10% (2% each)

Class participation will help your grade if you are on the border of a grade (e.g. B+ to A-). Class participation includes but is not limited to answering questions in class, attending office hours, helping others when appropriate in the lab, and serving in a leadership role in your project group.

Homework

Homework assignments will be posted on the Blackboard website **two or more weeks ahead of their due dates**. Homework is to be submitted outside of CILSE 403, via Gradescope, or in class **before** the beginning of the lecture (**1:30 sharp**) on the date specified. You can discuss your work **in the abstract** with other students in the class, but you **must** write up the solutions on your own.

Programs

In this class, you will have 3 programming assignments. **These will be done in groups of 2 students.** These programs can be written in any programming environment that you wish. More information will be provided on the programming assignments but the general areas are as follows:

Logic synthesis – The programming assignment will require you to enter a Boolean algebraic function(s), optimize it/them (via a variety of objective functions), and output the result in multiple formats.

FPGA Designer – This assignment will require that you create a virtual FPGA and “connect it” to the logic synthesis program you previously created. The program will assign the logic function to “LUTs” on the FGPA and connect those LUTs to realize the function as requested.

Physical Design – The final program will take your FGPA Designer tool, map it to a standard cell library, and perform a physical design activity of your choosing (partitioning, retiming, or placement/routing).

Programming descriptions will be posted on the Blackboard website **two or more weeks ahead of their due dates**. **Programs are due via GitHub links at 11:59 p.m. on the due date.** You may remain in the same group for the whole semester or change groups with each programming assignment. This is up to you. **Any partner conflicts should be reported EARLY to Prof. Densmore.**

Exams

There will be one midterm exam and a final exam. **The midterm will be TAKE HOME and you have 24 hours to return the exam. If you are unable to work on the exam during this period, you must provide 1 week advance notice along with appropriate documentation.** The final exam will be **TBD**. Make-up exams will only be given under extreme circumstances. NO AI on exams.

Project

Projects will be done in groups of **four students** (three and five-person groups will be allowed in extreme circumstances). **The project will represent a SIGNIFICANT amount of work and will culminate in a demonstrable digital system running on an FPGA OR an EDA-based programming tool.** There will be three project presentations and one demo session. The first two will be 10 minute, in-class updates. **Final project presentations will be during the last weeks classes and all team members must be present (see schedule).** Project presentations will be done in front of all students and should be treated as a professional presentation. More information regarding the projects will be provided during the semester. The project demo is the last day of class. Arrangements for the schedule and process of project demos will be made available later in the semester.

Previous project videos can be found at:

<https://www.youtube.com/channel/UCPYhfQlY30fTfEg7LpRPQtw>

Course Policy

- Homework/Programs: The homework assignments must be the result of your individual work (HW) or you and your partner (programs).
- You may discuss the contents and general approach to a problem with your classmates but not the detailed solution. You are expected to formulate your approach and to write the solutions to HW/Lab problems by yourself/group. Copying the solution and/or answer from another student is considered cheating. Two identical HWs/Labs with the same mistakes are considered cheating. **No extensions on homeworks or labs will be provided. If you will be absent on the day they are due you should make arrangements in advance with Prof. Densmore.**
- Makeup exams: Makeup exams will be provided if the student receives **prior permission** from the instructor. Emergencies will be dealt with on a case-by-case basis. Note that oversleeping, being not ready, or overload due to projects or coursework in other classes are **not** valid excuses for requesting a makeup exam.
- Exam/Home/Program Grade discussion: Grade discussion/corrections should be done within one week after the graded exam of homework is distributed. **No grade changes will be made after one week, or after the last day of class.**
- I and W grades: As per University policy.
- Honor Code: If you are found cheating on HWs, labs, or examinations, you will be brought up on charges before the Student Academic Conduct Committee whose punishment may include suspension from the University without the right to transfer credits for courses taken elsewhere.

AI Policy

- AI can NOT be used on Homework assignments in any form.
- AI can NOT be used on Exams in any form.
- AI should NOT be used for the paper discussions to CIRCUMVENT reading the paper or summarizing it. YOU CAN use it to check your summaries or to propose new questions for yourself or others. Only you will truly understand if you have read and understood the material.
- AI CAN be used on your programming assignments provided you cite your usage of it and are ready to explain your assignment if requested.
- AI SHOULD be used on your project to ensure your project is sufficiently complex, documented, usable, etc.
- If you are found to have violated this policy you may receive a “zero” on the associated assignment. In several cases, you may receive a failing grade for the course.
- If you have questions about the AI policy of the course, please contact Prof. Densmore.

Schedule for EC551 Fall 2026 - Lectures, Homework, Labs and Exams					
Lecture #	Date	Topic Description	Programs Out/Due	Hw Out	Hw Due
1	9/3 (Th)	Course Introduction			
2	9/8 (T)	EDA - Review of combinational logic		THQ1&2	
3	9/10 (Th)	EDA - Review of sequential logic			
4	9/15 (T)	EDA – Quiz review and grading			THQ1&2
5	9/17 (Th)	Verilog - Structural Verilog and Hierarchy	FGPA Tutorial	HW1	
6	9/22(T)	Verilog - Simulation and Test			
7	9/24 (Th)	RC – FPGAs + Paper 1 Out	Prog.1/Tut. (Fri)		
8	9/29 (T)	Verilog - Behavioral Verilog			
9	10/1 (Th)	Project - Plan and Goals Presentations		HW2	HW1
10	10/6 (T)	RC - Paper Review 1 + Paper 2 Out			
11	10/8 (Th)	EDA - Synthesis - Scheduling/Control			
12	10/15 (Th)	Midterm (take home)			
13	10/20 (T)	EDA - Synthesis – ASMs		HW3	HW2
14	10/22 (Th)	EDA - Synthesis - Multi-Level Logic	Prog.2/Prog.1 (Fri)		
15	10/27 (T)	RC - Paper Review 2 + Paper 3 Out			
16	10/29 (Th)	Project - Review and Update Presentations			
17	11/3 (T)	EDA - Static Timing Analysis		HW4	HW3
18	11/5 (Th)	Verilog - IP Cores and Microarchitecture	Prog.3/Prog.2 (Fri)		
19	11/10 (T)	RC - Paper Review 3 + Paper 4 Out			
20	11/12 (Th)	EDA - Retiming			
21	11/17 (T)	EDA - Design for Test/ Testing		HW5	HW4
22	11/19 (Th)	EDA – Physical Design	Prog.3 (Fri)		
23	11/24 (T)	RC - Paper Review 4			
		Thanksgiving Break			
24	12/1 (T)	Project - Final Presentations 1	Project Group1		
25	12/3 (Th)	Project - Final Presentations 2	Project Group2		HW5
26	12/8 (T)	Project Discussion and Class Survey			
	12/?? (?)	Final Exam (24 hour take home)			